# Climate Swings, Sticky Costs

## Year-over-Year Weather Volatility and the Financial Health of U.S. Counties, 2011–2023

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## MOTIVATION

Climate change is reshaping U.S. health finance not only through extreme events but through **year-to-year volatility**. Insurers reprice on new information, hospitals absorb uncompensated care, and households accumulate medical debt with a lag. Yet most economic studies estimate *level* effects of heat, drought, or pollution, treating a county with persistent exposure the same as one in transition.

**This poster asks:** does the *change* in exposure from one year to the next — a weather swing — independently burden the health-finance system, above and beyond the level?

If costs are driven by volatility, **adaptation planning must price in transitions, not just averages**.

## DATA

- **Panel:** 3,155 U.S. counties × 2011–2023 (41,376 county-years)
- **Climate** (NOAA National Centers for Environmental Information): divisional temperature, precipitation, Cooling Degree Days (CDD), Heating Degree Days (HDD), Palmer Drought Severity Index (PDSI); 1990–2000 baseline for county-specific z-scores
- **Air quality** (U.S. Environmental Protection Agency, EPA): annual Median and Max Air Quality Index (AQI) by county
- **Health-finance outcomes:** medical debt in collections (Urban Institute); Benchmark Silver premium (Health Insurance Exchange data, HIX Compare; rating-area → county); hospital bad debt per capita (National Academy for State Health Policy Hospital Cost Tool, NASHP HCT)
- **Economic outcomes:** Per-Capita Personal Income (Bureau of Economic Analysis, BEA); median household income and civilian employment (U.S. Census American Community Survey, ACS)

## EMPIRICAL STRATEGY

The primary specification isolates the marginal effect of a weather *swing* while controlling for the prior year's level:

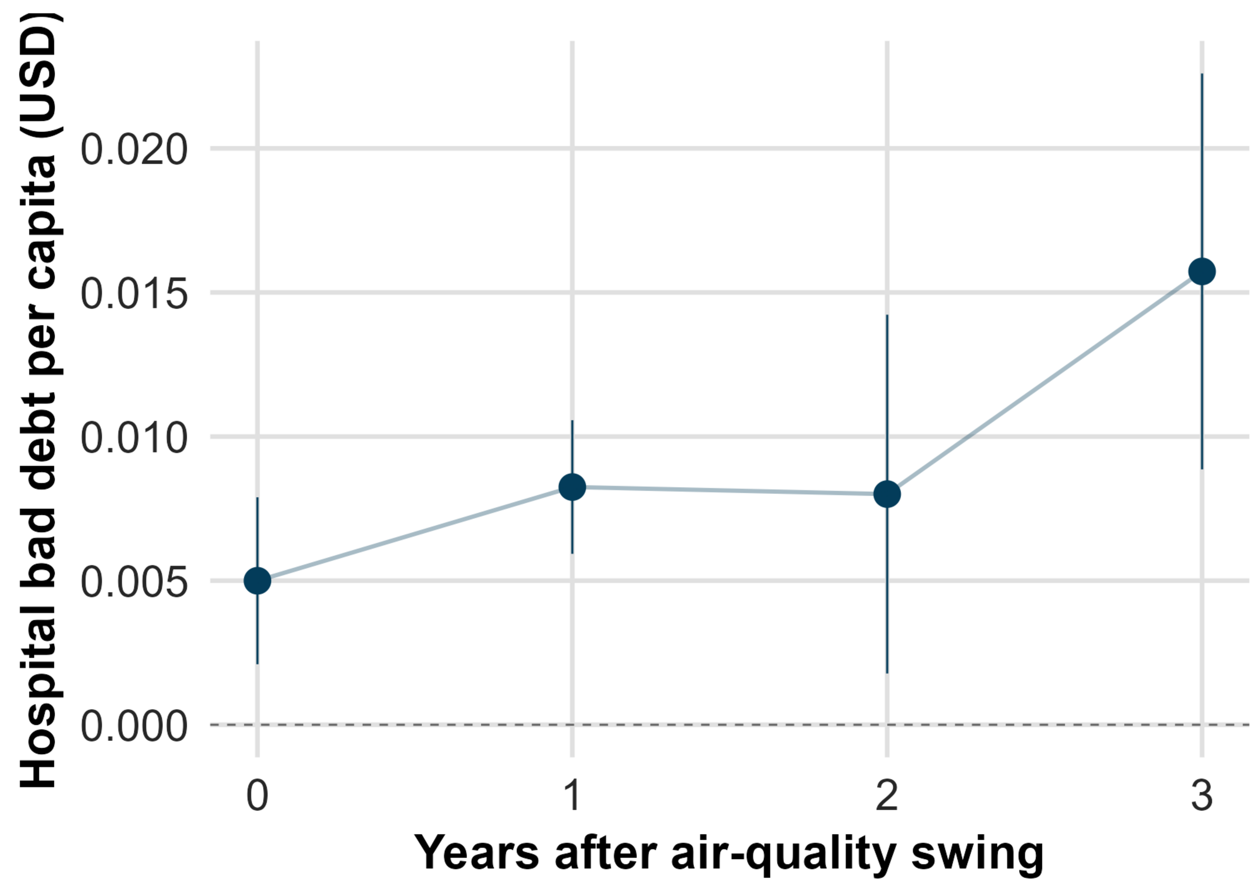
$$Y_{it} = \beta_1 \blacksquare X_{it} + \beta_2 X_{i,t-1} + \gamma \mathbf{Z}_{it} + \alpha_i + \tau_t + \varepsilon_{it}$$

- **$\beta_1$ (key):** effect of a one-unit change in exposure from $t-1$ to $t$, holding last year's level fixed. $\beta_2$ absorbs the residual level effect.
- $\alpha_i$, $\tau_t$: county FE absorb time-invariant heterogeneity (climate zone, baseline health, demographics); year FE absorb national shocks (inflation, CRA changes). $\mathbf{Z}_{it}$ controls: real PCPI, household income, employment.
- **Clustering:** state-level primary; rating-area for premium outcomes that share rating-area pricing.
- **Local projections** (Jordà 2005) extend to horizons $h = 0, 1, 2, 3$, tracing dynamic impulse-response without imposing lag form.
- **Decompositions:** asymmetric splits $\blacksquare X$ into positive (escalation) vs negative (relief); onset/exit/persist isolates regime transitions.
- **Exclusions:** CO 2023 dropped (HB23-1126) so a debt-reporting policy change is not read as a climate effect.

**896 coefficient estimates** across 5 specs × 7 outcomes × 7 exposures × 4 horizons.

## FINDING 1 — AQI SWINGS COMPOUND OVER TIME

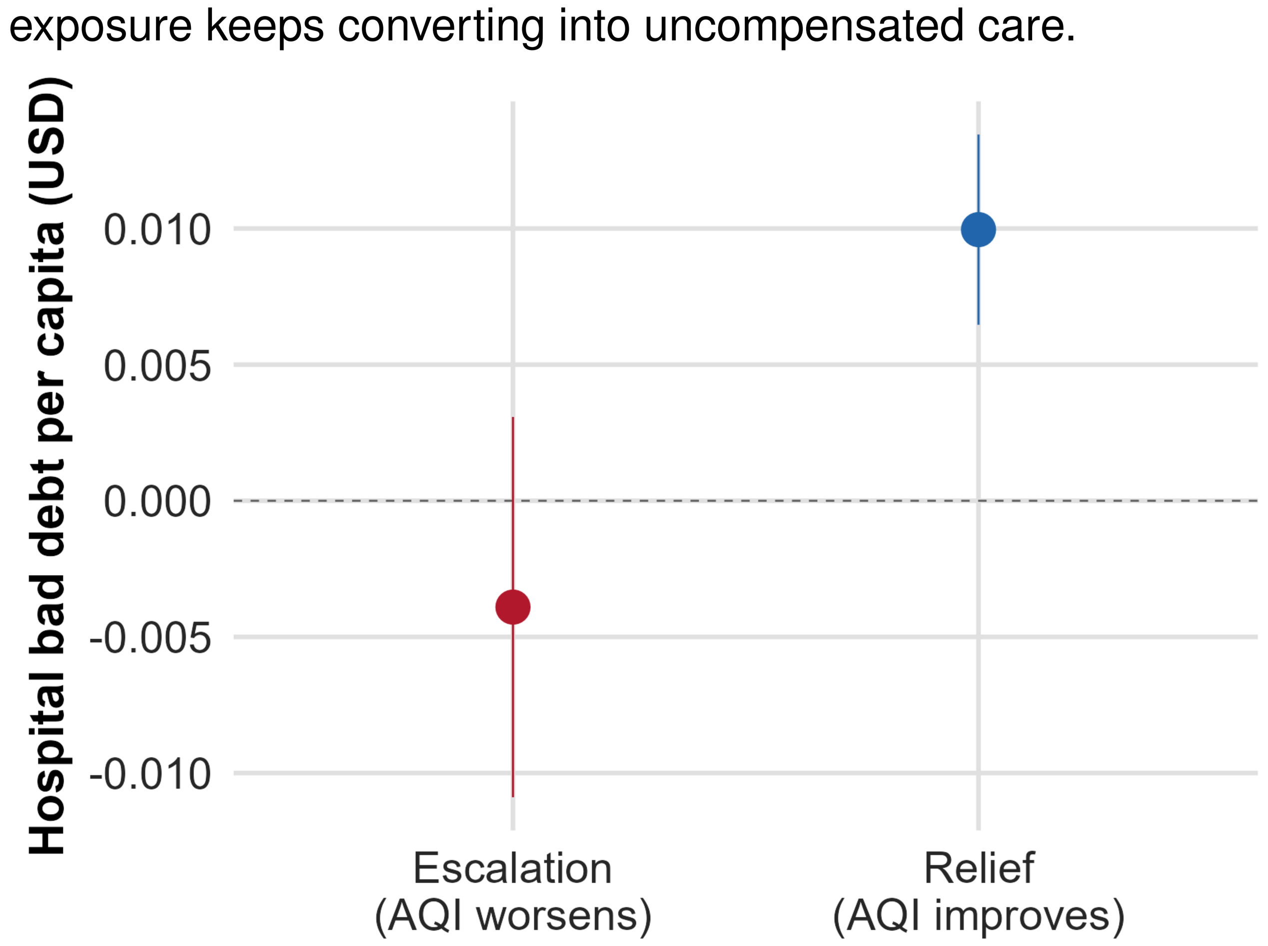
A one-unit positive swing in a county's Max AQI is associated with a **rising** stream of hospital bad debt across all four horizons, with the effect *growing* over time (h=0: +0.005, p=0.001; h=3: +0.016, p<.0001). There is no recovery — the cost accumulates.




**A worsening air-quality year today adds to hospital bad debt *three years later* — after claims mature, patients default, and losses are recognized.**

## FINDING 2 — THE RATCHET: RELIEF DOES NOT REVERSE

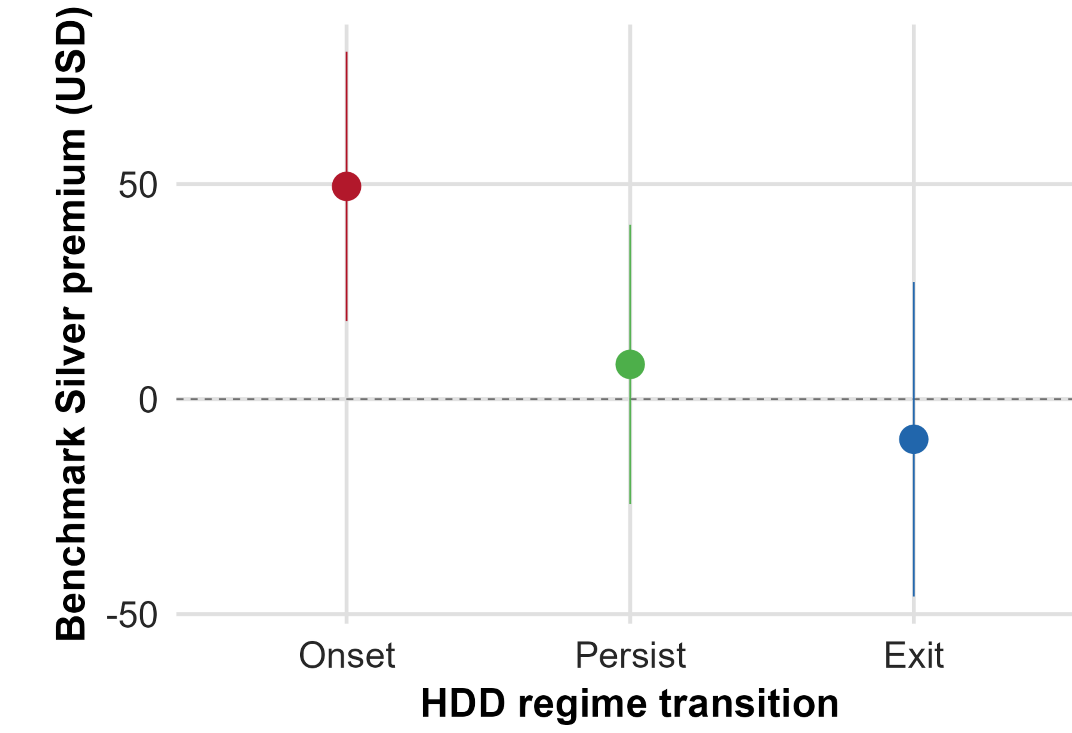
Splitting the swing by direction uncovers a counterintuitive pattern. **A year in which a county's air quality *improves* still raises hospital bad debt by +$0.010 per capita** (p<.0001). A year in which air quality *worsens* shows no significant contemporaneous effect. Bad debt climbs regardless of this year's trend — prior pollution exposure keeps converting into uncompensated care.




**The cost is a ratchet, not a thermostat.** Improving air quality does not unwind hospital bad debt within three years — prior exposure continues to mature into new uncompensated-care losses.

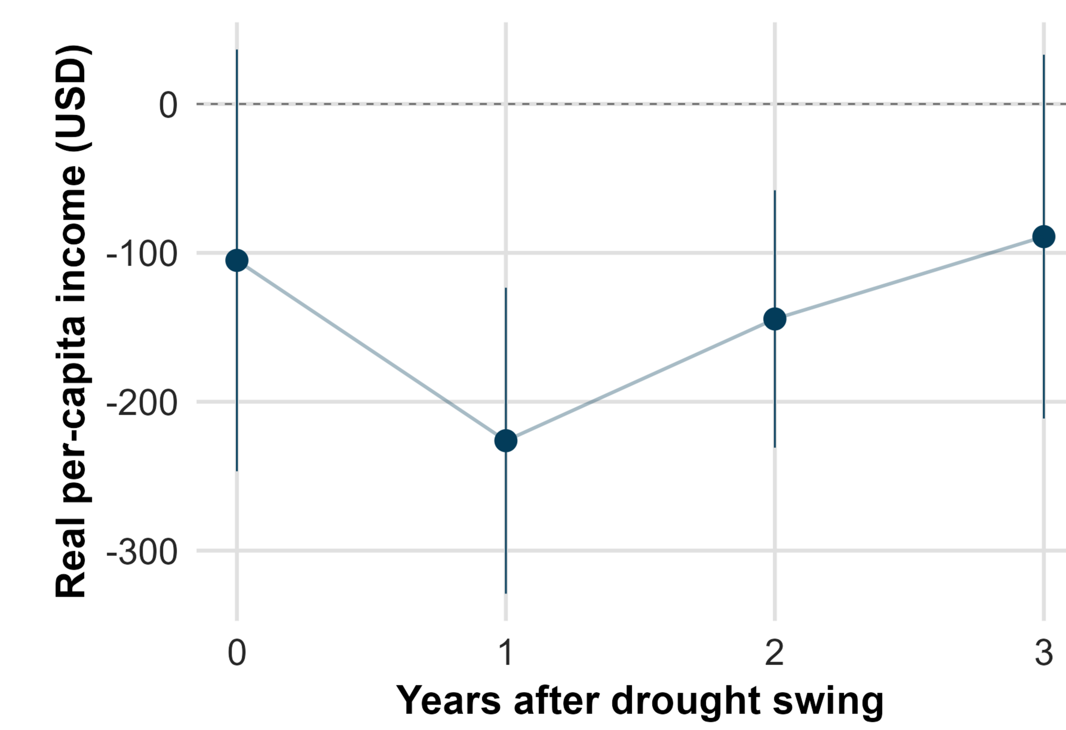
## FINDING 3 — COLD-SHOCK *ONSET* PRICES PREMIUMS

The first year a county *enters* a high-HDD regime: **+$49.46 on the Benchmark Silver premium** (p=0.003). Exit and persistence are null — insurers price at the transition, then hold.




## FINDING 4 — DROUGHT SWINGS SUPPRESS INCOME

A worsening PDSI swing cuts per-capita personal income for two years (h=1: −$226, p<.0001; h=2: −$144, p=0.002); concentrated in **drought-worsening swings** (−$207, p=0.001), with employment loss (−142 jobs, p=0.006).




## TAKEAWAYS

- **Volatility is its own exposure.** The swing channel is significant where the level channel was not (AQI → hospital bad debt).
- **Costs ratchet, not reverse.** Pollution relief does not refund hospital bad debt within three years; insurers reprice on shock *onset*, not exit.
- **Drought volatility hits the macro base.** Income and employment contract for multiple years, feeding back into medical-debt formation.

**Acknowledgments.** Thanks to faculty **Avi Dor**, **Susan Anenberg**, **Eric Luo**, and **Cindy Liu** for guidance on specification, data, and interpretation.